

CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

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Membrane Builder

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Membrane Builder helps the user generate a series of CHARMM inputs necessary to build a protein/membrane complex for molecular dynamics simulations. A brief description of each step is given below. Among various other building schemes, either the "insertion" or the "replacement" method can be chosen by the user in step 3. (user can choose one of them in step 3, see below).

- Insertion method

A protein is inserted into a pre-equilibrated lipid bilayer with a hole whose size is comparable to the protein size (the libraries of lipid bilayers are available in [archive](#))

- Replacement method

A protein is first packed by lipid-like spheres whose positions are subsequently used to place randomly chosen lipid molecules from the library (the libraries of lipid molecules are available in [archive](#))

Please note that

- If you are not familiar with Membrane Builder, please first watch these [video demos](#) and also read the relevant references below.
- **NAMD inputs (v2.7b3 or after)** are provided for equilibration and production (see [STEP6](#)). Input files can be found in "namd" directory when you download tar archive ("charmm-gui.tgz") after all the input file generation.
- **GROMACS inputs (v5.0 or after)** are provided for minimization, equilibration, and production (see [STEP6](#)). Input files can be found in "gromacs" directory when you download tar archive ("charmm-gui.tgz") after all the input file generation. See gromacs/README.
- **AMBER inputs (v16 or after)** are provided for minimization, equilibration, and production (see [STEP6](#)). Input files can be found in "amber" directory when you download tar archive ("charmm-gui.tgz") after all the input file generation. See amber/README.
- **GENESIS inputs (v1.1.0 or after)** are provided for minimization, equilibration, and production (see [STEP6](#)). Input files can be found in "genesis" directory when you download tar archive ("charmm-gui.tgz") after all the input file generation.
- **OpenMM inputs (v6.2 or after)** and running scripts are provided for equilibration and production (see [STEP6](#)). Input files can be found in "openmm" directory when you download tar archive ("charmm-gui.tgz") after all the input file generation. See openmm/README.
- **CHARMM/OpenMM inputs (c39b1 or after)** are provided for equilibration and production (see [STEP6](#)). Input files can be found in "charmm_openmm" directory when you download tar archive ("charmm-gui.tgz") after all the input file generation.
- **TINKER inputs (v8.10 or after)** are provided for minimization, equilibration, and production (see [STEP6](#)). Input files can be found in "tinker" directory when you download tar archive ("charmm-gui.tgz") after all the input file generation.
- the protein must be oriented with respect to a membrane bilayer whose normal is parallel to the Z-axis and whose center is located at Z=0
- RCSB PDB structures are NOT pre-oriented, but can be oriented in STEP 2 (see below)
- Users can use Download Source OPM (<http://opm.phar.umich.edu>) that provides pre-oriented protein coordinates with respect to the membrane normal
- The OPM PDB does not contain "TER" between ATOM and HETATM, so that CHARMM-GUI often fails to recognize ligand molecules. In such case, the user should manually insert "TER" in appropriate places. In addition, all carbohydrate connection information is lost in OPM PDB files.
- Users can use Download Source RCSB and then use PPM web server (http://opm.phar.umich.edu/ppm_server) in STEP 2 to obtain a oriented protein coordinate with

respect to the membrane normal, so that users can have all molecular information (disulfide bonds / ligands / carbohydrates etc) in a RCSB PDB file

- a homogeneous lipid bilayer can be built with DMPC, DPPC, DOPC, POPC, DLPE, and POPE
- a heterogeneous lipid bilayer can be built with 434 different lipid molecules (see [lipid list](#))
- the heterogenous Membrane Builder can be used for a homogeneous lipid bilayer (only using the replacement method)
- P21 crystal image is available in CHARMM input option
- rectangular and hexagonal geometries are available for a system shape in XY
- If you are not familiar with the first PDB reading step, please first watch these [video demos](#).

References for Membrane Builder:

S. Jo, T. Kim, V.G. Iyer, and W. Im (2008)

CHARMM-GUI: A Web-based Graphical User Interface for CHARMM. [J. Comput. Chem. 29:1859-1865](#)

S. Jo, T. Kim, and W. Im (2007)

Automated Builder and Database of Protein/Membrane Complexes for Molecular Dynamics Simulations. [PLoS ONE 2\(9\):e880](#)

S. Jo, J.B. Lim, J.B. Klauda, and W. Im (2009)

CHARMM-GUI Membrane Builder for Mixed Bilayers and Its Application to Yeast Membranes. [Biophys. J. 97:50-58](#)

E.L. Wu, X. Cheng, S. Jo, H. Rui, K.C. Song, E.M. Dávila-Contreras, Y. Qi, J. Lee, V. Monje-Galvan, R.M. Venable, J.B. Klauda, and W. Im (2014)

CHARMM-GUI Membrane Builder Toward Realistic Biological Membrane Simulations. [J. Comput. Chem. 35:1997-2004](#)

J. Lee, X. Cheng, J.M. Swails, M.S. Yeom, P.K. Eastman, J.A. Lemkul, S. Wei, J. Buckner, J.C. Jeong, Y. Qi, S. Jo, V.S. Pande, D.A. Case, C.L. Brooks III, A.D. MacKerell Jr, J.B. Klauda, and W. Im (2016)

CHARMM-GUI Input Generator for NAMD, GROMACS, AMBER, OpenMM, and CHARMM/OpenMM Simulations using the CHARMM36 Additive Force Field. [J. Chem. Theory Comput. 12:405-413](#)

J. Lee, D.S. Patel, J. Stähle, S.-J. Park, N.R. Kern, S. Kim, J. Lee, X. Cheng, M.A. Valvano, O. Holst, Y. Knirel, Y. Qi, S. Jo, J.B. Klauda, G. Widmalm, and W. Im (2019)

CHARMM-GUI Membrane Builder for Complex Biological Membrane Simulations with Glycolipids and Lipoglycans. [J. Chem. Theory Comput. 15:775-786](#)

J. Lee, M. Hitzenberger, M. Rieger, N.R. Kern, M. Zacharias, and W. Im (2020)

CHARMM-GUI supports the Amber force fields. [J. Chem. Phys. 153:035103](#)

S. Park, Y.K. Choi, S. Kim, J. Lee, and W. Im (2021)

CHARMM-GUI Membrane Builder for Lipid Nanoparticles with Ionizable Cationic Lipids and PEGylated Lipids. [J. Chem. Inf. Model. 61:5192-5202](#)

☒ Protein/Membrane System

Download PDB File:
Download Source: RCSB ▼

Upload PDB File: Choose File 8hbv_with_...rdiolipins.pdb

PDB Format: ☒ PDB ☐ PDBx/mmCIF ☐ CHARMM

☐ Check/Correct PDB Format 

☐ [Membrane Only System](#)

References for Lipid Force Fields:

J.B. Klauda, R.M. Venable, J.A. Freites, J.W. O'Connor, D.J. Tobias, C. Mondragon-Ramirez, I. Vorobyov, A.D. MacKerell, Jr., and R. W. Pastor (2010)

Update of the CHARMM all-atom additive force field for lipids: Validation on six lipid types [J Phys Chem B. 114\(23\): 7830-7843](#)

R.M. Venable, A.J. Sodt, B. Rogaski, H. Rui, E. Hatcher, A.D. MacKerell, Jr., R.W. Pastor, and J.B. Klauda. (2014) CHARMM All-Atom Additive Force Field for Sphingomyelin: Elucidation of Hydrogen Bonding and of Positive Curvature. [Biophys J. 107\(1\):134-45](#)

Next Step: 
Select Model/Chain

A brief explanation of each step:

- **STEP1: *Read protein coordinates***
The user can download the coordinates from RCSB (PDB website) or OPM (<http://opm.phar.umich.edu>). OPM provides pre-oriented protein coordinates with respect to the membrane normal. The user can upload the PDB (or CHARMM) format coordinates from the user's local machine, once you properly orient the protein in membranes.
- **STEP2: *Orient the protein***
If the PDB coordinates are from RCSB, it is necessary to properly orient the protein with respect to membranes. There are two options for doing this. It is the step in which the cross-section area of the protein along the Z-axis is calculated and displayed. The maximum top ($10 < Z < 20$) and bottom ($-20 < Z < -10$) areas are used to determine the system size in XY.
For a detailed example showing how to orient your structure, see [this](#) video demo, particularly the part beginning at 2:00.
- **STEP3: *Determine the system size***
To determine the system size in XY, there are three options, in the case of homogeneous bilayer generation, based on (1) number of lipid layers around the protein, (2) specific number of lipid molecules in top and bottom, and (3) specific size of the system along X and Y. The system size along Z is determined by the water extent from the top and bottom of the protein. For now, two types of system shapes in XY (rectangular and hexagonal) are available.

In the case of heterogeneous bilayer generation, there are two options to determine the system size: (1) ratio of lipid types to be used and initial (guess) size of the system along X and Y (2) specific number of lipid molecules and ratio of system size along X and Y. The system size along Z is determined by the water extent from the top and bottom of the protein. If desired, hydration number (number of water molecules per one lipid molecule) can be used for this purpose.
- **STEP4: *Build the components***
Based on the system size determined in the previous step, this step builds individual pieces such as (1) the lipid bilayer around the protein, (2) additional water molecules to fully solvate the protein, and (3) ions (with Monte Carlo sampling or distance-based algorithm) for a given concentration.
- **STEP5: *Assemble the components***
All the pieces (protein, lipid bilayer, additional water, and ions) are assembled together in this step.
- **STEP6: *Equilibrate the system***
Due to their computation time, only the input files for six "suggested" equilibration steps are provided. However, the user can find equilibrated coordinates for some membrane proteins from the archive.

NAMD inputs are provided for equilibration (step6.1 through step6.6) and production

- `namd/step6.[1-6]_equilibration.inp`: use generated coordinates of a membrane/protein complex (or membrane only system) to perform equilibration. Equilibration inputs use collective variable restraints to slowly release the system to facilitate stable simulation.
- `namd/step7.1_production.inp`: use the restart file from NAMD and continue production runs.

GROMACS inputs are provided for minimization (step6.0), equilibration (step6.1 through step6.6) and production

- `gromacs/step6.0_minimization.mdp`: use generated coordinates of a membrane/protein complex (or membrane only system) to perform minimization.
- `gromacs/step6.[1-6]_equilibration.mdp`: use minimized coordinates from GROMACS to perform equilibration.
- `gromacs/step7_production.mdp`: use equilibrated coordinates from GROMACS and continue production runs.

AMBER inputs are provided for minimization (step6.0), equilibration (step6.1 through step6.6) and

production

- amber/step6.0_minimization.mdin: use generated coordinates of a membrane/protein complex (or membrane only system) to perform minimization.
- amber/step6.[1-6]_equilibration.mdin: use minimized coordinates from AMBER to perform equilibration.
- amber/step7_production.mdin: use equilibrated coordinates from AMBER and continue production runs.

GENESIS inputs are provided for minimization (step6.0), equilibration (step6.1 through step6.6) and production

- genesis/step6.0_minimization.inp: use generated coordinates of a membrane/protein complex (or membrane only system) to perform minimization.
- genesis/step6.[1-6]_equilibration.inp: use minimized coordinates from GENESIS to perform equilibration.
- genesis/step7_production.inp: use equilibrated coordinates from GENESIS and continue production runs.

OpenMM inputs and running scripts are provided for equilibration (step6.1 through step6.6) and production

- openmm/openmm_run.py: OpenMM running scripts written in python.
- openmm/step6.[1-6]_equilibration.inp: use generated coordinates of a membrane/protein complex (or membrane only system) to perform equilibration.
- openmm/step7_production.inp: use the restart file from OpenMM and continue production runs.

CHARMM/OpenMM inputs are provided for equilibration (step6.1 through step6.6) and production

- charmm_openmm/step6.[1-6]_equilibration.inp: use generated coordinates of a membrane/protein complex (or membrane only system) to perform equilibration.
- charmm_openmm/step7_production.inp: use the restart file from CHARMM/OpenMM and continue production runs.

TINKER inputs are provided for minimization, equilibration, and production

- tinkers/step6_minimization.key: use generated coordinates of a membrane/protein complex (or membrane only system) to perform minimization.
- tinkers/step6_equilibration.key: use minimized coordinates from TINKER to perform equilibration.
- tinkers/step7_production.key: use the restart file from TINKER and continue production runs.

